Enabling Power Meter Reading Applications

MG301 Series Metering Gateway

- LTE CAT 1/4
- Rich Interfaces
- IEC62056-21
- DLMS/COSEM
- Li-ion Battery

1. Product Overview

MG301 is a wireless metering gateway designed according to the requirements of the Automatic Meter Reading System for communication via cellular networks. MG301 enables the server to access the connected metering devices via the cellular network, allowing the server to obtain the operational parameters of the meters at any time and record detailed traceability information. MG301 is equipped with various hardware interfaces such as Ethernet port, serial port, digital input, relay output, etc., enabling it to connect to various meters on-site and achieve multiple applications.

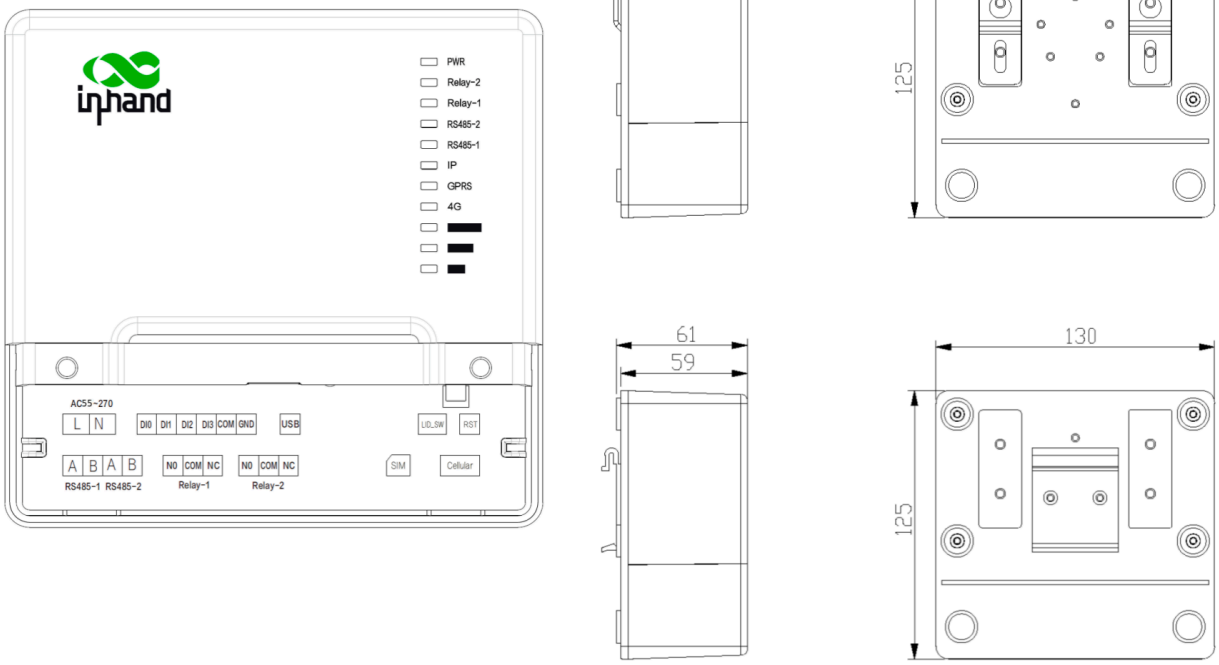
Key Technical Specifications

Specification	Details
Cellular	4G cellular module, CAT1/CAT4
Hardware Platform	MIPS24KEc (575/580 MHz); 128MB RAM; 64MB Flash
Metering Protocols	IEC62056-21 Mode-C; DLMS/COSEM; Modbus RTU
Platform Access	MQTT/MQTTS/MASS/TeleMetriX Head-End
Interfaces	2 × isolated RS485; 1 × USB2.0 Type-C; 4 × DI; 2 × DO; 2 × latching relay outputs (optional); 1 × 100Mbps Ethernet (optional)
Power	55–270 VAC, nominal 230 V; 20 VA (max.)
Dimensions	130 × 125 × 68 mm (specification table)
Operating Environment	-30 to 70 °C; 0–93% ambient humidity
Protection	IP40

2. Features and Advantages

Feature	Description
Stable Communication	2G/4G (LTE CAT 1)/NB-IoT (optional); independent hardware and software watchdog; multilevel link detection and communication self-healing
Rich Interfaces	2 × RS-485; 1 × USB2.0; 2 × latching relay outputs; 4 × DI + 2 × DO; 1 × 10/100Mbps Ethernet (optional)
Performance and Platform	MIPS24KEc (575/580 MHz); 128MB RAM; 64MB Flash
Multiple Protocols	IEC62056-21; DLMS/COSEM; Modbus RTU; MQTT/MQTTS; MASS; TeleMetriX Head-End; FTP
Industrial-grade Design	Operating temperature -30 to 70 °C; +55 to 270 VAC power input; IP40; Li-ion battery or supercapacitor

3. Product Dimensions



Dimension (specification table): 130 × 125 × 61 mm.

4. Product Interfaces

Interface	Description
Ethernet	1 × 100Mbps Ethernet port, RJ45 connector (optional)
Cellular	4G cellular module, CAT1/CAT4
USB	1 × USB2.0, Type-C
Serial Port	2 × RS485, isolated, terminal-block connector; RS485: Data+, Data-, GND
SIM Card	Mini SIM-2FF card, supports 1.8/3V SIM/UIM card; eSIM
Antenna Connector	1 × Cellular Antenna
Button	1 × System reset button
Open Lid Detection	1 × Open Cover Inspection key
Relay Outputs	2 × relay outputs: 240VAC/5A, latching relay output (optional)
Digital Inputs	4 × digital inputs; isolation voltage: 3500 V; dry contact: Logic 1, Open; Logic 0, Close to ground; wet contact: VIH(max.) = 30 VDC, VIH(min.) = 5 VDC, VIL(max.) = 2 VDC
Digital Outputs	2 × digital outputs; isolation voltage: 3500 V; supply voltage: 20 V max.; sink current: 0.5 A max.

LED Indicators



LED	Status Description
PWR	Steady on: The power supply is normal Off: No external power supply is available
Relay-2	Steady on: Closing
Relay-1	Steady on: Closing
RS485-2	Blinking: Data is being exchanged through serial port 2 Off: No data is being exchanged over serial port 2
RS485-1	Blinking: Data is being exchanged through serial port 1 Off: No data is being exchanged over serial port 1
IP	Steady on: The device obtains an IP address for network communication Off: The device does not obtain an IP address for network communication
GPRS	Steady on: 2G Off: Not 2G
4G	Steady on: 4G Off: Not 4G

5. Hardware Specifications

Category / Parameter	MG301
Hardware Platform	
CPU	MIPS24KEc (575/580 MHz)
RAM	128MB
FLASH	64MB
Interfaces	
Ethernet Port	1 × 100Mbps Ethernet port, RJ45 connector (optional)
Cellular	4G cellular module, CAT1/CAT4
USB	1 × USB2.0, Type-C
Serial Port	2 × RS485, isolated, terminal-block connector; RS485: Data+, Data-, GND
SIM Card	Mini SIM-2FF, supports 1.8/3V SIM/UIM card; eSIM
Antenna Connector	1 × Cellular Antenna
Button	1 × System reset button
Open Lid Detection Device	1 × Open Cover Inspection key
Relay Outputs	2 × relay outputs: 240VAC/5A, latching relay output (optional)
Digital Inputs	4 × digital inputs; isolation voltage: 3500 V; dry contact: Logic 1, Open; Logic 0, Close to ground; wet contact: VIH(max.) = 30 VDC, VIH(min.) = 5 VDC, VIL(max.) = 2 VDC
Digital Outputs	2 × digital outputs; isolation voltage: 3500 V; supply voltage: 20 V max.; sink current: 0.5 A max.
Other Hardware	

Category / Parameter	MG301
RTC	Real time clock operated from on-board coin-cell battery
Punctuality Accuracy	0.5 S/day
Battery	Li-ion battery (500mAh) or supercapacitor (Li-ion battery default)
Power	
Power Supply	AC input: unregulated 55V to 270 VAC, nominal 230V, instantaneous voltage protection of 6KV; reverse voltage protection
Power Consumption	20VA (max.)
Mechanical Characteristics	
Mounting	DIN rail / mounting ear (default)
Enclosure Material	Plastic
Dimension	130 × 125 × 61 mm
Protection	IP40
Ambient Temperature and Humidity	
Operating Temperature	-30 to 70 °C
Ambient Humidity	0–93%
Declared Pollution Degree	3

Category / Parameter	MG301
Overvoltage Category	CAT III
LED	
System	1 × Power, 3 × CSQ, 1 × IP, 1 × 4G, 1 × GPRS, 2 × RS485, 2 × Relay
Safety & EMC	
Safety & EMC	IEC 62368-1; EN 55032 Class B; EN 55035 Level 3
Other Standards	
Others	EN 60068-2-30; EN 60068-2-31; EN 60068-2-1; EN 60068-2-2; EN 62262; IEC 60529; IEC 60695-11-10 standard V-0 class

6. Software Specifications

Parameter	Specification
Operating System	
OS	Linux
Internet	
Access Authentication	APN, VPDN
Internet Access	CHAP, PAP
Network Format	LTE CAT 1
Network Protocol	
IP Application	ICMP, DNS, TCP/UDP, TCP Server, DHCP, NTP

Parameter	Specification
IP Routing	Static routing
Reliability	
Cover Opening Alarm	Detection lid is opened
Power Failure Alarm	Detecting an external power failure
Link Detection	Sends heartbeat packets to detect, auto redials when disconnected
Embedded Watchdog	Device runs self-detection, auto recovers from malfunctions
Temperature Detection	The CPU temperature can be detected and recorded in logs
Communication Protocol	
Acquisition Protocol	IEC62056-21 Mode-C; DLMS/COSEM; Modbus RTU
Access Platform	MQTT/MQTTS/MASS/TeleMetriX Head-End
Network Management	
Configuration	Web via USB/ETH
Upgrade	Local via USB/ETH; remote via FTP/InHand DM platform
Log	Support local system log and important log power-off autosave

7. Ordering Information

Model Code

Model: MG301-FQ53 / MG301-FQ33 / MG301-FQ68

Product Models

Model	Cellular Type & Frequency Band	Serial Port	USB	I/O	Ethernet (Optional)
MG301-FQ53	EMEA, LTE CAT1 FDD: B1/B3/B5/B7/B8/B20/B28 TDD: B38/B40/B41 GSM: B2/B3/B5/B8	2 × RS485	1 × USB2.0	2 × relay outputs, 4 × DI, 2 × DO	1 × 100Mbps Ethernet
MG301-FQ33	North America, LTE CAT1 FDD: B2/B4/B5/B12/B13/B25/B26 WCDMA: B2/B4/B6	2 × RS485	1 × USB2.0	2 × relay outputs, 4 × DI, 2 × DO	1 × 100Mbps Ethernet
MG301-FQ68	LATAM, LTE CAT4 FDD: B1/B2/B3/B4/B5/B7/B8/B28 TDD: B40 WCDMA: B1/B2/B5/B8 GSM: B2/B3/B5/B8	2 × RS485	1 × USB2.0	2 × relay outputs, 4 × DI, 2 × DO	1 × 100Mbps Ethernet

8. Contact Us

- **Website:** InHand Networks
- **Copyright:** © InHand Networks. All rights reserved.